PixelSmile: Toward Fine-Grained Facial Expression Editing

Jiabin Hua, Hengyuan Xu, Aojie Li, Wei Cheng, Gang Yu, Xingjun Ma et al. | Fudan University

https://arxiv.org/abs/2603.25728

Research Questions

- How can we address semantic overlap and confusion between similar facial expressions (e.g., fear vs. surprise, anger vs. disgust) in fine-grained expression editing?
- How can we enable continuous and linearly controllable expression intensity editing while preserving identity consistency?
- How can we systematically evaluate expression editing models across multiple dimensions including structural confusion, controllability, and identity preservation?

Core Idea

The core problem is that facial expressions naturally overlap on a continuous semantic manifold - fear and surprise share wide eyes and open mouth, while anger and disgust both involve frowning and negative affect. This semantic overlap causes systematic confusion in both recognition and generation models when trained with discrete one-hot labels. The key insight is to replace discrete supervision with continuous 12-dimensional affective scores and use fully symmetric joint training to explicitly contrast confusing expression pairs. This forces the model to learn disentangled representations that can distinguish between semantically similar expressions while enabling smooth interpolation for controllable intensity editing.

Key Formula

Textual Latent Interpolation

$$e_{cond}(\alpha) = e_{neu} + \alpha \cdot \Delta e, \quad \alpha \in [0, 1]$$

e_neu is neutral prompt embedding, e_tgt is target expression embedding, Δe = e_tgt - e_neu is the residual direction, α controls expression intensity. Enables continuous control by interpolating in textual latent space.

Symmetric Contrastive Loss

$$L_{SC} = \frac{1}{2}[T(G_a, P_a, P_b) + T(G_b, P_b, P_a)]$$

G_a, G_b are generated images for expressions a,b; P_a, P_b are ground truth positive/negative pairs; T is triplet constraint function. Symmetric design avoids directional bias by swapping positive/negative roles to disentangle confusing expressions.

Key Figure

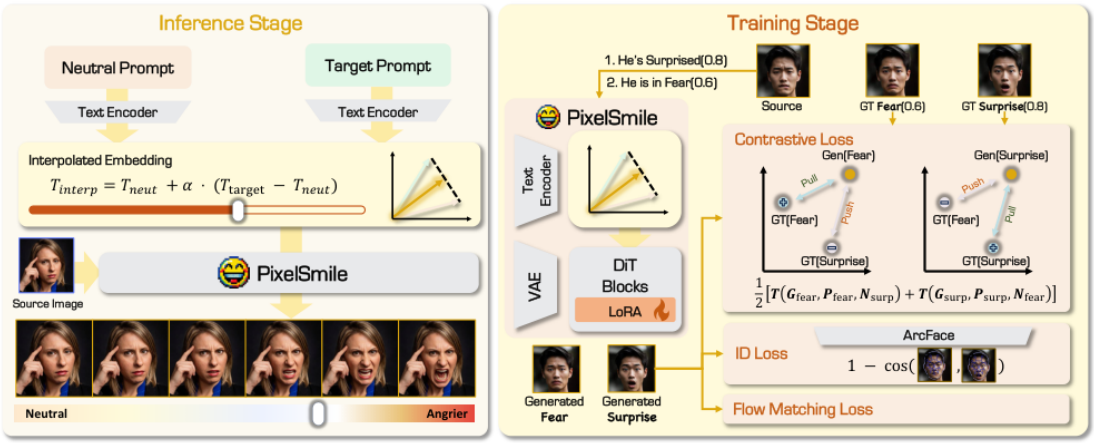

Figure 3. **Framework Overview.** (1) **Inference Stage**. We interpolate between the neutral and target expression embeddings in textual latent space using a controllable coefficient α, enabling continuous adjustment of expression intensity. (2) **Training Stage**. We adopt a joint fully symmetric training framework. Specifically, we sample a source image P_{src} and a confusing expression pair (P_a, P_b) to construct a triplet. We first treat P_a as the positive and P_b as the negative to compute a joint loss, and then swap their roles to compute it again, yielding a symmetric training objective. The joint loss consists of three components: a Flow-Matching loss for intensity alignment, a contrastive loss for expression separation, and an identity preservation loss to maintain subject consistency.

Figure 3: Shows the complete PixelSmile framework with inference stage (textual latent interpolation) and training stage (symmetric joint training with triplet loss), illustrating the core technical approach.

Limitations

- Some expression combinations produce unstable results due to physiological conflicts (e.g., Angry+Happy)
- Performance depends on VLM scoring model bias - highly reliable on basic expressions but less consistent on extended categories
- Dataset inherits demographic biases from portrait-centric internet images with non-uniform distributions

- Expression blending only works for 9 out of 15 combinations, with some pairs collapsing to single dominant expression